Tunisian Administrative Procedures Assistant

A Local, Source-Grounded Multi-Agent RAG System for Multilingual Administrative Guidance

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Abstract

Information on Tunisian administrative procedures is spread across ministry web pages, the National Business Register (RNE) site and Official Journal (JORT) PDFs, in French and Arabic, while many citizens ask questions in Tunisian Derja. Large language models answer such questions fluently but may invent fees, deadlines or required documents. We present a fully local assistant that answers in English, French, Arabic and Derja (Arabic and Latin script) exclusively from 13 verified official documents (12 indexed). Five agents orchestrated by n8n route the question, retrieve evidence from Qdrant, quote official checklists verbatim, generate citation-checked answers, and detect empty fields in photographed official forms. On a 84-question evaluation set whose relevance labels are verified verbatim against the official texts, a hybrid retriever (dense BGE-M3 on raw and expanded queries plus BM25, reciprocal rank fusion and a per-document cap) raises Hit@5 from 0.704 to 0.873 and MRR@5 from 0.577 to 0.675 over the Phase-2 agent retriever, and from 0.000 to 0.714 for Latin-script Derja on a parallel multilingual subset, at a median retrieval cost of 386 ms. On 20 held-out questions written after all tuning, the same retriever reaches Hit@5 0.750 (Phase-2 retriever: 0.562; raw dense retrieval: 0.750), showing that part of the development gain was overfitting.

Keywords: retrieval-augmented generation, multilingual retrieval, Tunisian Arabic, multi-agent systems, n8n, hallucination mitigation, document analysis.

1 Introduction and problem statement

Obtaining a passport or registering a company in Tunisia requires combining several official sources: the Ministry of the Interior publishes procedure guides in French and Arabic [7]; the RNE publishes a table of 95 procedures with fees and legal deadlines [8] and the legal texts that govern them [10, 11]. These sources are fragmented, not always consistent (we found that the French and Arabic passport pages state different stamp duties, 20/60 vs. 25/80 TND), and written in languages that differ from the one many users speak (Derja, often typed in Latin script with digits, e.g. *"9adech ya5ou wa9t el passbor?"*).

A general-purpose chatbot is a risky answer to this problem: hallucinated requirements or amounts [4] can cost citizens time and money. Our objectives were therefore: (i) answer only from verifiable official documents, with citations; (ii) refuse when the corpus has no answer; (iii) support four query languages without altering official text; (iv) check official forms for missing fields; (v) run entirely on local hardware (one laptop GPU with 4 GB VRAM) without paid APIs or exposure of personal data. Scope: passports, the national identity card and RNE business registration. Birth certificates are *deliberately not covered*: the only source found (idaraty.tn) opts out of AI use in its `robots.txt`, so the system reports the procedure as uncovered.

2 System architecture

Figure 1 shows the deployed system (Docker Compose, all ports bound to `127.0.0.1`). A Telegram bridge uses Bot API long polling, so no public URL is needed, and forwards each message to an n8n webhook; HTTP clients use the same webhook. n8n orchestrates five agents exposed by a FastAPI backend. Qdrant stores the vectors; Ollama on the host GPU serves BGE-M3 embeddings [1], Mistral 7B [5] (routing fallback and grounded answers) and `translategemma:4b` (French-to-English translation of quoted items). All processing is local.

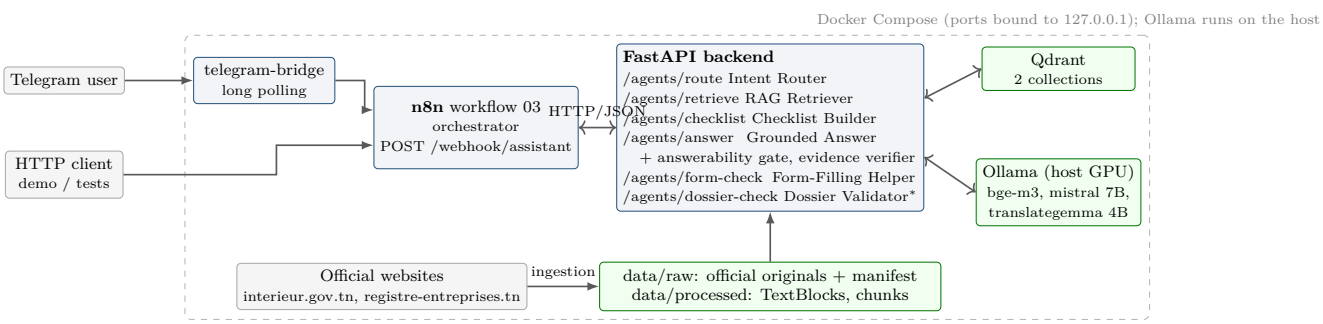


Figure 1: Implemented architecture. Arrows are HTTP/JSON calls; green boxes are stores and model servers. *Innovation extension, reached through workflow 03 with `mode: dossier`.

Technology choices. *RAG instead of fine-tuning* [6]: the knowledge must remain traceable to a source document and updatable when a decree changes, and fine-tuning a 7B model is not feasible on 4 GB VRAM. *n8n* makes

the agent chain visible and editable as a workflow and provides scheduling for re-ingestion. *Qdrant* offers payload filtering on procedure category and stores the full provenance with each chunk. *Ollama* keeps user questions and uploaded documents on the machine.

3 Official corpus and RAG pipeline

Provenance. A source registry (`config/sources.yaml`) lists every candidate with its status: 13 approved documents (passport guides FR/AR, national ID guide, the RNE formalities table, RNE law 2018-52, decree 2022-298 on fees, decree-law 2022-18, two decrees of 2019, an order of 2026, two official forms and a scanned decree), plus excluded and not-found candidates with the reason; the passport application form was not downloaded because the server's `robots.txt` disallows it. The downloader enforces `robots.txt`, AI opt-out directives and TLS verification; each original is stored unmodified with its SHA-256, retrieval time and HTTP metadata in a manifest. Official references and publication dates are copied from the documents themselves; legal applicability stays `unverified` until a human confirms it.

Extraction and chunking. PDFs (PyMuPDF) and HTML pages are converted to typed *TextBlocks* (heading, paragraph, list item, table row) with page numbers; Arabic presentation forms are normalised with NFKC. Two chunkers consume the same blocks (Table 2): a *structure-aware* chunker that keeps headings, list sections and table rows together (max. 1200 characters, heading path prefixed) and a *recursive* character splitter (1000 characters, 150 overlap). Chunks are embedded with BGE-M3 (1024-d, cosine) into two Qdrant collections with the full provenance as payload. The scanned decree has no text layer and is skipped rather than OCR'd into the public corpus.

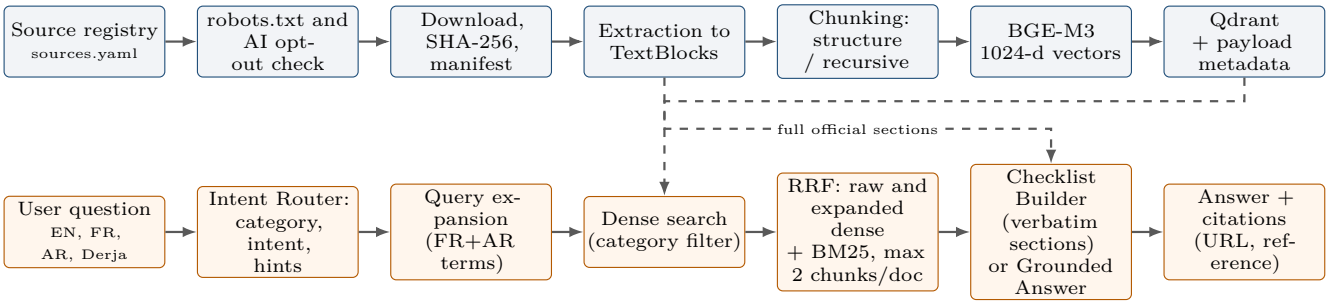


Figure 2: Ingestion and indexing (top) and query-time retrieval (bottom). Dashed arrows: stored artefacts used at query time.

Retrieval (Fig. 2). The Phase-2 retriever expanded the user query with fixed bilingual search terms for the routed procedure and intent (e.g. the French and Arabic headings for "documents to provide"), filtered on the procedure category and gave a small bonus to chunks in the user's script. Section 6 shows that this expansion helps transliterated Derja but drowns specific questions in the 95 near-identical rows of the RNE table. The final retriever therefore fuses, with reciprocal rank fusion (RRF, $k = 60$) [2], three lists restricted to the routed category: dense retrieval of the raw query, dense retrieval of the expanded query (both embedded in one batched call), and BM25 [9] over the raw query plus deterministic term normalisation (company types such as *public limited company* → *Société anonyme*; common Arabizi words such as *khtiya* → the Arabic-script word for "fine"); at most two chunks per document enter the top 5. **Citations** are not generated: they are copied from the chunk payload (title, organisation, URL, official reference, page, SHA-256).

4 Multi-agent system and prompt engineering

Figure 3 shows workflow 03. The design principle is *deterministic first*: an LLM is used only where no rule can do the job, and every LLM output is validated.

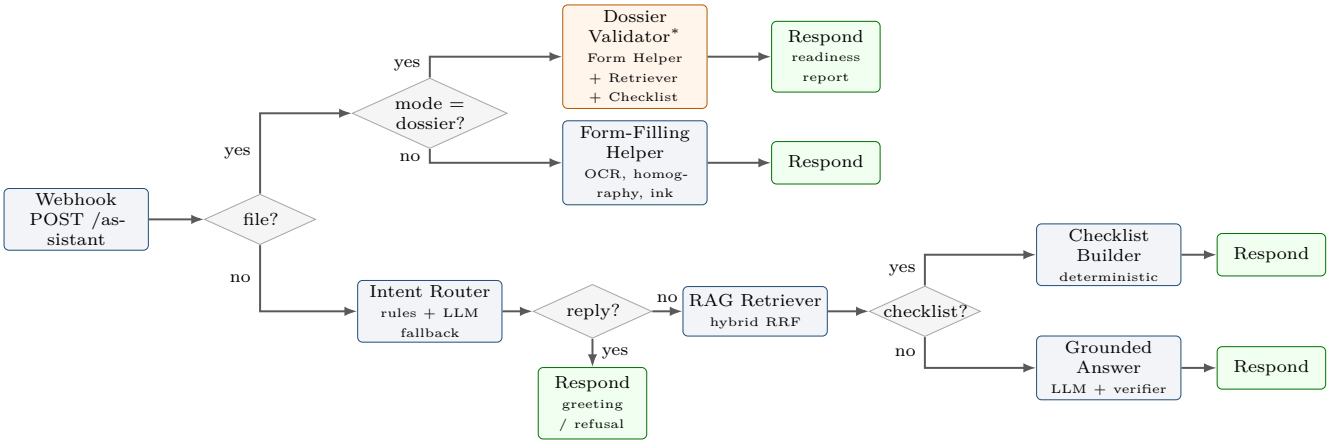


Figure 3: n8n workflow 03 as deployed (16 nodes): each box is an HTTP node calling one backend agent; diamonds are n8n IF nodes. *Uploads with `mode: dossier`; uploads without it keep the unchanged form check.

- **Intent Router.** Language detection (EN/FR/AR/Derja, including Latin-script Derja via digit-letters such as *3, 7, 9*) and keyword rules give the procedure, intent and hints (abroad, loss, renewal). The LLM prompt `intent_router@v1` is called only when no procedure keyword matches; its JSON output is checked against fixed enumerations. RNE questions go to the Checklist Builder only when they name a procedure code or a registration act (company type, registration, transfer, ...); other fee or deadline questions (fee decree, name reservation) go to the Grounded Answer agent.
- **RAG Retriever.** Section 3. Unknown procedures are searched without filter.
- **Checklist Builder.** Retrieval only identifies the document and the *scenario* (e.g. passport, renewal, resident abroad). The complete official sections (conditions, documents, place, deadline) are then read verbatim from the extracted structure, so chunk boundaries never truncate a list. When the same guide exists in French and Arabic with identical structure, Arabic and Derja users receive the Arabic text, and the amounts of both versions are compared: a discrepancy produces a warning. Translation is applied only French→English and is rejected unless every line and every number is preserved. For the RNE, where no official checklist exists, the agent says so and quotes the official table rows in retrieval order (plus top-ranked legal passages verbatim); a requested procedure code absent from the table (e.g. RNE C 014) is reported as non-existent instead of being answered with other rows.
- **Grounded Answer.** For free questions (penalties, beneficial owners, ...) Mistral 7B answers from numbered passages; answers without citations or citing non-existent passages are rejected and the passages are quoted instead. An *answerability gate* runs first: if the procedure was not identified by the keyword rules, at least half of the question's IDF-weighted informative terms (words absent from the corpus count as maximally informative) must occur in the retrieved passages, otherwise the assistant refuses instead of answering or quoting unrelated evidence. An *evidence verifier* then checks (i) topic consistency (Section 7) and (ii) every number written with digits in the LLM answer against the passages cited by the same sentence, reading source numbers both as digits and as French, English or Arabic number words; sentences with unsupported numbers are removed, and if no cited sentence remains the answer is refused (no number is ever replaced).
- **Form-Filling Helper.** Section 5.

Versioned prompts. Prompts live in `prompts/<name>/<version>.yaml`; a registry selects the active version, and every LLM response records its `prompt_ref`. Iterations (Table 6): (1) `grounded_answer` v1→v2: an Arabic answer was cut by the 600-token limit; v2 raised it to 900 and asked to reproduce official lists. Measured on the 12 questions routed to this agent, v2 raised acceptance (valid citations) from 0.667 to 0.750 and response-language consistency from 0.917 to 1.000, but verbatim evidence in the answer fell from 3/11 to 1/11, as the model paraphrased more. (2) v3 therefore requires quoting the official words before explaining them: evidence rose only from 1/11 to 2/11, acceptance was unchanged and the median time grew from 24.7 to 31.7 s. With one item of difference on 11, v3 was *not adopted*; v2 remains active. (3) `translate_items@v1` FR→AR was *withdrawn* rather than re-prompted: it passed format checks but changed meanings (*veuve*, widow, became "one-eyed woman"); Arabic and Derja users receive the official text with a note. Numbers in all accepted LLM answers occurred in the passages given to the model (faithfulness 1.0 in every run).

5 n8n automation and multimodality

Workflow 01 runs ingestion and indexing (manual or weekly); workflow 02 exposes plain retrieval; workflow 03 (Fig. 3) answers users and is configured not to store execution data, because executions would contain user questions and files. The bridge splits long answers at line boundaries (Telegram limit 4096 characters) and never logs message contents or the bot token.

Form analysis. Field templates are derived from the official blank forms RNE F-001 and F-002: every dot-leader line is a field labelled by the adjacent French and Arabic text. Digital PDFs are checked geometrically (words not printed on the blank form lying on a dot leader). For photos and scans, the page is deskewed by a projection-profile search, OCR'd with Tesseract `ara+fra` [12], aligned to the blank form by a RANSAC homography [3] over anchor words, and the ink in each dot-leader band is compared with the blank form rendered at the same scale. OCR text alone proved unusable for this decision: text written on dot leaders is read merged with the dots at near-zero confidence. Uploads are processed in memory only and never indexed. Speech input is not implemented.

Innovation: Smart Administrative Dossier Validator. `POST /agents/dossier-check`, reached through workflow 03 when an upload carries `mode: dossier` (Fig. 3), extends the Form-Filling Helper into a readiness report for the official RNE forms F-001 and F-002, keeping three notions apart. *Form-field status*: each detected field is `filled`, `missing` or `unreadable`; signature and checkbox sections are `cannot_verify`. *Official requirements*: only statements an official source makes, quoted verbatim with their status (`supported_by_official_source`, `not_verified`, `official_requirement_unavailable`): the instructions printed on the form itself (all data must be completed, a missing item leads to rejection, e-mail and mobile number are mandatory, a declaration lacking the required documents is void) and the matching rows of the RNE formalities table (fee, deadline). *Dossier completeness* is never certified: the validator asks the RAG Retriever and Checklist Builder whether an official list of supporting documents exists for the procedure; for the RNE none does, so the report states that completeness cannot be assessed instead of reconstructing a list with an LLM. The validator is deterministic (no LLM call), reports field statuses but never field values, and summarises in the user's language. 17 tests (official blank forms and placeholder-filled test fixtures; recognition, rejection of unrelated files, filled/missing/unreadable fields, verbatim official quotes, absence of validation claims, privacy of values, four languages) pass, and the OCR path returned the exact field statuses in 6/6 scan-like images (both forms; blank, two fields filled, rotated 2°). It was not evaluated on real handwriting or photographs.

6 Evaluation methodology

Dataset. 84 questions (34 FR, 26 EN, 11 AR, 13 Derja of which 9 in Latin script) covering passports (25), the national ID (4), RNE registration, fees, forms and legal texts (45), birth certificates (4, unsupported) and out-of-scope requests (5). Ground truth is never written by hand: each answerable item names *evidence spans* that must occur verbatim in a named official document; a build script verifies every span against the extracted text and stores the page, the full official block and the source URL. 69 items have verified evidence, 2 ambiguous items have document-level labels, 13 have no answer in the corpus. Equivalent French and Arabic passages form one evidence *group*; conflicting items have one group per version. A *parallel subset* asks 7 verified facts (and one unsupported request) in 4–5 language/script variants each. Human review of the labels is pending.

Metrics. A retrieved chunk is relevant if it belongs to the group's document and contains the span (after Unicode, whitespace and diacritic normalisation). Recall@k is the share of groups found in the top k, Hit@5 whether any group is found, MRR@5 the reciprocal rank of the first relevant chunk. All configurations use the same questions, routes (cached from the production router), filters and $k = 5$. Answer quality is measured on the full production chain with deterministic checks only (no LLM judge): routing accuracy, citation validity against the manifest (URL and SHA-256), presence of a verified evidence span, faithfulness (checklist items verbatim in the official text; numbers in LLM answers present in the passages given to the model), refusal on unsupported items (refusal-phrase heuristic for LLM answers), conflict warnings and response script. Latencies were measured on the development laptop (RTX 3050 4 GB, host-side calls through `127.0.0.1: localhost` added a 2.1 s IPv6 fallback per request on this Windows host, an artefact we found and excluded).

7 Results and failure analysis

Table 1: Retrieval on the 71 questions with verified relevance labels ($k = 5$, median latency per query).

Configuration	Chunks	R@1	R@3	R@5	MRR@5	Hit@5	ms
Dense, raw query (Phase 1)	structure	0.465	0.676	0.718	0.575	0.732	368
Dense, raw query (Phase 1)	recursive	0.528	0.725	0.803	0.642	0.817	360
Agent retriever (Phase 2, before)	structure	0.493	0.648	0.697	0.577	0.704	–
Agent + term normalisation	structure	0.507	0.662	0.711	0.591	0.718	364
Agent + term normalisation	recursive	0.415	0.556	0.648	0.504	0.662	366
Hybrid RRF (raw+expanded+BM25)	structure	0.549	0.768	0.831	0.668	0.845	393
Hybrid RRF + max 2 chunks/doc	structure	**0.549**	**0.782**	**0.859**	**0.675**	**0.873**	386
Hybrid RRF + max 2 chunks/doc	recursive	0.507	0.711	0.768	0.613	0.775	396
Hybrid RRF + max 3 chunks/doc	structure	0.549	0.768	0.852	0.669	0.859	388

Retrieval and chunking (Tables 1, 2). With raw dense retrieval, recursive chunks beat structure-aware chunks (Hit@5 0.817 vs. 0.732): the 95 structure chunks of the RNE table are near-identical and, at 430 characters median, too short to carry distinctive context. With routing, fusion and the per-document cap the ranking reverses: structure-aware chunks give the best configuration (Hit@5 0.873, MRR@5 0.675) because one table row is exactly one chunk, which BM25 can match on exact codes (*RNE M 004.03*) and entity names, while recursive chunks merge several rows. Each component was kept only after a measured gain: fusion +0.127 Hit@5 over the agent retriever, the cap +0.028, term normalisation +0.014 for the agent retriever; a cap of 3 was worse than 2. Batched embedding keeps the hybrid within 22 ms of plain dense retrieval.

Table 2: Chunking statistics.

Strategy	Chunks	Mean chars	Median chars
structure	272	526	430
recursive	174	867	911

Table 3: Hit@5 by query language/script (all 71 items).

Subset	n	Dense	Agent	Hybrid+cap
ar	9	0.778	0.778	1.000
derja	4	0.750	1.000	1.000
derja-latin	8	0.125	0.750	0.750
en	21	0.810	0.619	0.857
fr	29	0.828	0.724	0.862

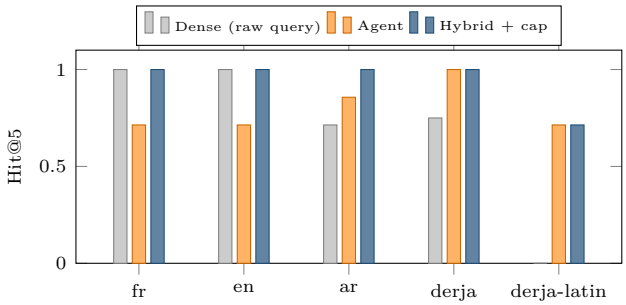


Figure 4: Parallel subset: the same 7 facts asked in each language (Derja in Arabic and Latin script).

Multilingual retrieval (Fig. 4, Table 3). BGE-M3 handles French, English and Arabic questions over French and Arabic documents well, but raw dense retrieval finds *none* of the Latin-script Derja questions of the parallel subset: words such as *passbor*, *charika* or *khtiya* have no counterpart in either script. Routed expansion and the Arabizi map raise this to 0.714; the residual failures (beneficial-owner and penalty questions in Arabizi) use vocabulary outside the map. Because the map was compiled after observing failures and no held-out set exists, this gain is likely optimistic. Official passages are never translated for retrieval; only queries are expanded.

Remaining retrieval failures. (i) English or French questions about the fee decree (refunds, payment) still lose to RNE table rows; (ii) R23 asks in French about an appeal deadline that exists only in the Arabic law, ranked below the 2026 order; (iii) *which form* questions (R32) are poorly served because the forms' text is mostly field labels.

Table 4: Answer quality of the production chain (84 items).

Check (deterministic)	Rate	n
Category routing accuracy	0.952	83
Intent routing accuracy	0.819	83
Language detection accuracy	1.000	84
Citation validity (URL and SHA-256 match manifest)	1.000	145
Citation support (an expected document is cited)	0.855	69
Verified evidence span present in response	0.565	69
Faithfulness (verbatim items / numbers in sources)	1.000	65
Unsupported / out-of-scope rejection	1.000	12
Source conflict flagged	1.000	2
Response language consistency (strict, whole text)	0.821	84

Table 5: End-to-end latency by final agent.

Final agent	n	Median ms
checklist builder	49	428
grounded answer	22	26163
intent router	13	2304

Table 6: Prompt iteration on the questions routed to the Grounded Answer agent (same passages for both versions).

Run	Prompt	Accepted	Numbers in sources	Evidence span	Language OK	Median s
A	grounded_answer@v1	0.667	1.000	0.273	0.917	29.2
A	grounded_answer@v2	0.750	1.000	0.091	1.000	26.5
B	grounded_answer@v2	0.750	1.000	0.091	1.000	24.7
B	grounded_answer@v3	0.750	1.000	0.182	1.000	31.7

Answer quality (Table 4). The final chain (hybrid retrieval, prompt v2) cites only real manifest entries (147/147 citations match URL and SHA-256), never quotes an item that is not in the official text, and copies every number of an accepted LLM answer from its passages (faithfulness 1.0). Both conflicting fee questions carry the FR/AR warning. Error analysis on the first full runs led to fixes, each re-measured on the same 84 items: unknown procedure codes are reported as non-existent; RNE fee or duration questions without a registration act go to the Grounded Answer agent instead of the table; special scenarios (pilgrimage, change of passport) are no longer chosen for generic questions; and the answerability gate refuses unsupported topics. Rejection of unsupported requests rose from 0.833 to 0.917 and then to 1.000 (the driving-licence question O04 is now refused), and citation support from 0.826 to 0.855. Verbatim evidence is 0.565, a strict lower bound: R25 is answered correctly in French (*valable pour une durée de six mois, renouvelable une fois*) from an Arabic-only decree, and English checklists are validated

translations, so neither can contain the official span verbatim.

Remaining failures. (i) Routing: 4 category errors, all from the LLM fallback (e.g. R28 uses the indefinite Arabic form of "beneficial owner", which the keyword rules do not cover, and was sent out of scope); intent accuracy is 0.819, mostly fee/duration/information confusions that do not change the agent. (ii) Questions asking *which* RNE form to use get the generic greeting rather than the form links. (iii) Strict language consistency (0.821) fails mainly by design: English and Derja users receive verbatim French originals when no validated translation or Arabic version exists. Excluding verbatim official quotations, the assistant's own text is in the user's language in 0.929 of the answers.

Held-out validation (Table 7). Because every fix above was found on the development set, 20 new questions (16 with verified evidence, 4 unsupported) were written on official facts not used during development and evaluated once before and once after the last changes (answerability gate; localised labels before verbatim official lines; a fraction-slash bug that made every translation of the national-ID photo requirement fail validation; Derja detection), without tuning on them. Retrieval does not generalise as well as on the development set: the production hybrid ties raw dense retrieval on Hit@5 (0.750) with a lower MRR@5 (0.531 vs. 0.635), but still clearly beats the Phase-2 retriever (0.562). Citations stay 100 % valid; language detection reaches 1.0, strict language consistency rises from 0.70 to 0.80 and assistant-text consistency from 0.944 to 1.000. Two failures remain unseen during development: (H17) "carte grise" is answered with national-ID evidence, because the generic words *renouveler* and *carte* give the gate 58 % lexical support; (H14) an accepted LLM answer contains a number absent from its passages (faithfulness 0.938). These two discovered failures motivated the evidence verifier below; from then on the held-out set is no longer independent for them, and new test sets were written.

Table 7: Development vs. held-out results before and after the final reliability changes (retrieval was not changed; production configuration).

	Development (84)		Held-out (20)	
Metric	before	after	before	after
Hit@5, dense raw query	0.732		0.750	
Hit@5, Phase-2 agent retriever	0.718		0.562	
Hit@5, hybrid + cap (production)	0.873		0.750	
MRR@5, hybrid + cap (production)	0.675		0.531	
Citation validity	147/147	145/145	31/31	31/31
Unsupported-answer rejection	0.917	1.000	0.750	0.750
Faithfulness	1.000	1.000	0.938	0.938
Language consistency, strict	0.798	0.821	0.700	0.800
Language consistency, assistant text	0.909	0.929	0.944	1.000
Verified evidence span present	0.580	0.565	0.375	0.312

Evidence verification (Table 8). H14 and H17 had two root causes: the answer agent only checked that citation markers existed, so the LLM could convert the Arabic "fifty thousand dinars" of Law 2018-52 into "510,000 TND"; and nothing checked that the evidence described the procedure asked about. The numerical check is deterministic (median 3 ms) and removed every unsupported number on all five sets, including two independent test sets written before it was built, and never caused a false refusal by itself (the refusals it triggers replace answers whose figures were wrong). Topic verification was harder. A first version asked Mistral 7B (prompt `evidence_topic@v1`, 6.6 s median per call) whether the evidence describes the asked procedure; its verdict proved unstable (the same "carte grise" question was judged consistent in one run and inconsistent in another, depending on the excerpts shown). The retained version is deterministic for French and Arabic questions, the corpus languages: a topical content word that occurs in none of the official documents (after light stemming; verb forms excluded) means the subject is not covered; English and Derja questions keep the LLM verdict. It removes the known failure type (regression set: false acceptance 0.143→0) but, on the independent test sets, it did *not* reduce false acceptance and added false refusals: synonyms absent from the corpus ("tarif" instead of "redevance") and word forms ("fixe") are read as uncovered subjects, while uncovered subjects named with corpus words ("attestation de travail", a supporting document only) still pass. On test set 2, 5 of 6 answerable questions were already refused by the router's LLM fallback without the verifier. The topic check is therefore *disabled by default* and kept for experiments (`EVIDENCE_TOPIC_CHECK`); the numerical check is a clear gain and stays enabled.

Table 8: Without (off) and with (on) the evidence verifier. [a]Contaminated: H17 and H09 were analysed before the final version. [b]Second evaluation (first: verifier v1). [c]Untouched, evaluated once with the final verifier; "off" is the same code with the verifier disabled.

	False acceptance		False refusal		Unsupp. numbers		Valid citations	
Set	off	on	off	on	off	on	off	on
Development (84)	0.000	0.000	0.058	0.058	2/85	0/78	145/145	145/145
Regression (15)	0.143	0.000	0.000	0.125	1/9	0/1	14/14	11/11
Held-out (20)[a]	0.250	0.000	0.000	0.125	1/9	0/1	31/31	25/25
Test 1 (14)[b]	0.250	0.250	0.000	0.100	2/19	0/17	23/23	19/19
Test 2 (12)[c]	0.167	0.167	0.833	1.000	2/7	0/8	2/2	1/1

Other component tests. 227 unit tests and 20 integration tests pass (official forms, extraction, chunking, routing, checklist construction on the real corpus, metrics). Integration tests run the n8n chain end to end. The Telegram bridge was tested with a *mocked* Telegram API forwarding to the real n8n chain; a live Telegram conversation is *not* part of this evaluation. The form helper recognised the form and returned exactly the set of empty fields in 30/30 synthetic image cases (two official forms; blank, partial and complete fills with placeholder text; 120–200 dpi, JPEG, 1–3° rotation, perspective). Real handwriting and real photographs were not tested because no real filled forms were used.

8 Security, limitations and conclusion

Privacy and security. All models run locally; services listen on `127.0.0.1`; the Telegram bridge needs no inbound port. User questions, uploads and the bot token are never logged; uploads are decoded in memory, never written to disk and never indexed; n8n stores no executions of workflow 03. Users are advised to send redacted forms.

Limitations. The corpus is small (13 documents) and its legal currency is unverified: publication in the JORT does not prove that a text is still in force, and the two passport pages disagree. There is no official RNE checklist of required documents; the system states this instead of guessing. Retrieval is weakest for Latin-script Derja and for fee-decree questions; the evaluation sets were built by the developers and their labels await human review; the held-out set is small (20 items), and it shows that part of the development gain was overfitting. The lexical answerability gate can be fooled by generic shared words (H17), and the topic verifier trades it for false refusals on synonyms; neither guarantees legal correctness. LLM paths are slow on 4 GB VRAM (Mistral 7B runs 40 % on GPU), translation is restricted to French→English, and speech input, birth certificates and the passport application form are not covered.

Future work. Human review and a larger held-out set; a stronger semantic topic verifier (local entailment model or multilingual concept lexicon) and a better routing fallback; more official sources (birth certificates with the operator's permission); a learned Arabizi transliterator; reranking once hardware allows; and a live Telegram field test.

Conclusion. Grounding every administrative statement in verbatim official text, measuring retrieval against verified evidence rather than model judgement, and refusing when the corpus is silent produced an assistant whose answers can be traced to their source. The evaluation also showed where intuition failed: the production query expansion lowered Hit@5 for specific questions, and the better chunking strategy depended on the retriever. Both were found only because every change was measured on the same verified questions.

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